

学习笔记四

一. md 文件图片插入问题

1.在GitHub上传了我上次的md文件，其中的照片的确无法显示。

打开Windows Powershell，输入指令下载deepseek

 [图片-20260428194544689](#)

4.在pycharm里接入deepseek

 [image-20260428194828323](#)

原因：

打开Windows Powershell，输入指令下载deepseek

![图片-20260428194544689](C:\Users\35641\AppData\Roaming\Typora\typora-user-images\image-20260428194544689.png)

4.在pycharm里接入deepseek

![image-20260428194828323](C:\Users\35641\AppData\Roaming\Typora\typora-user-images\image-20260428194828323.png)

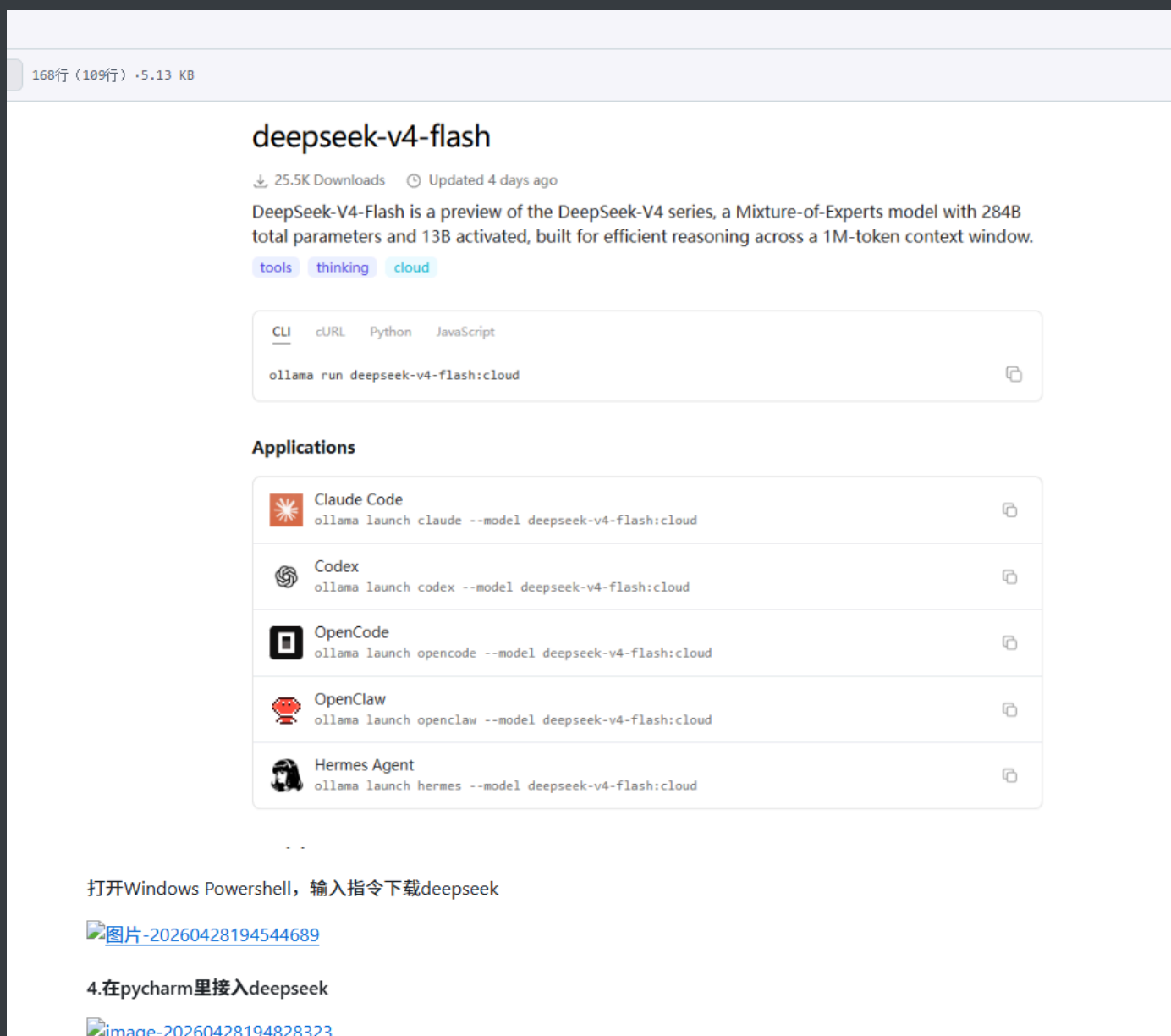
- 原代码中使用了本地绝对路径，上传到GitHub中无法访问。
- 路径中用了反斜杠“\”，GitHub只能识别“/”正斜杠。原因：windows系统用反斜杠来作为路径分隔，Linux系统使用正斜杠，github服务器运行在Linux系统。

2.解决方法：

- 创建一个文件夹，用于存储写在GitHub上的图片。
- 把md文件里的图片代码这样写

 ![image-20260428194109043](image/image-20260428194109043.png)

- 在上传到github上的时候，把md文档和文件夹一起上传



如图所示修改了代码的图片可以正常显示未修改的图片依然无法显示。

二、第三方package：IPython.display, bs4, requests

1. IPython.display

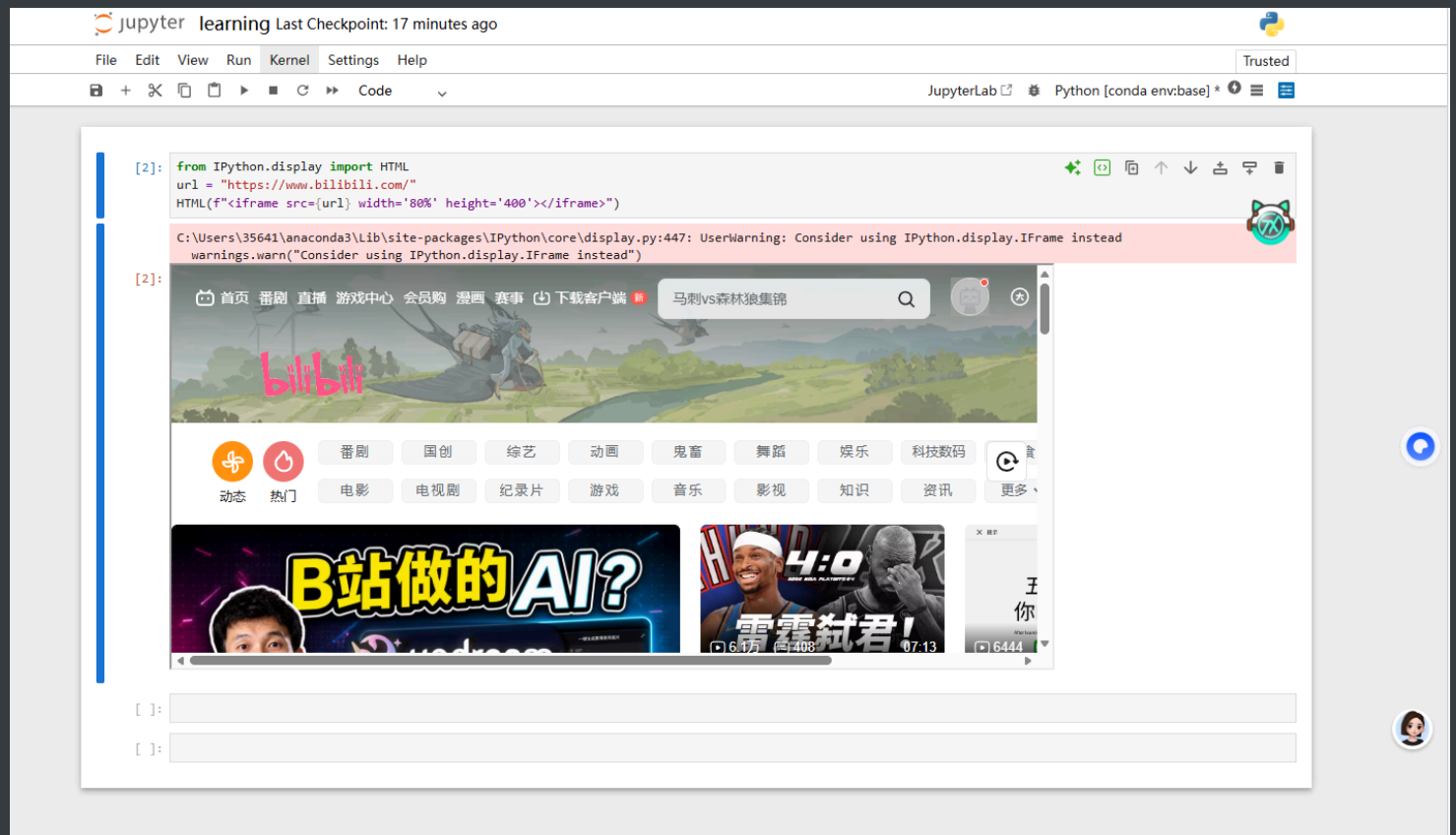
这是一个在Jupyter可以用来展示各种格式的内容的工具

例如：

```
from IPython.display import HTML
url = "网址"
HTML(f"<iframe src={url} width='80%' height='400'></iframe>")
```

#<iframe>:HTML中的结束标签 src={url}嵌入某个网站 width='80%': 宽度占当前容器80%
height='400': 400像素 HTML: 把文本当作HTML代码渲染。

运行实例如下：



2.bs4, requests

bs4可以用于在乱七八糟的网页中找到需要的东西。

- res.text:网页的源代码
- find_all ()：“帮我找出所有...” (返回列表)
- find ()：找出第一个
- get_text：只得到文字不得到标签

requests用于得到网页上的东西。

```

from bs4 import BeautifulSoup
import requests
from IPython.display import HTML, display
url = "https://httpbin.org/html"
res = requests.get(url) #获取这个网址的内容
soup = BeautifulSoup(res.text, "html.parser") #res.text指的是获取的网址源代码，html.parser
是HTML解析器，用来帮助程序看懂网页结构
all_text = soup.find_all("p") #找到带有“p”这个标签的段落
combined_text = "" #创建一个空的字符串
for text in all_text:
    combined_text = combined_text + "\n" + text.get_text() #把所有的段落拼起来
print(combined_text)

```

运行如下：

```

[12]: !pip install bs4

Requirement already satisfied: bs4 in c:\users\35641\anaconda3\lib\site-packages (0.0.2)
Requirement already satisfied: beautifulsoup4 in c:\users\35641\anaconda3\lib\site-packages (from bs4) (4.13.5)
Requirement already satisfied: soupsieve>1.2 in c:\users\35641\anaconda3\lib\site-packages (from beautifulsoup4->bs4) (2.5)
Requirement already satisfied: typing-extensions>=4.0.0 in c:\users\35641\anaconda3\lib\site-packages (from beautifulsoup4->bs4) (4.15.0)

```

```

[13]: from bs4 import BeautifulSoup
import requests
from IPython.display import HTML, display
url = "https://httpbin.org/html"
res = requests.get(url)
soup = BeautifulSoup(res.text, "html.parser")
all_text = soup.find_all("p")
combined_text = ""
for text in all_text:
    combined_text = combined_text + "\n" + text.get_text()
print(combined_text)

```



Availing himself of the mild, summer-cool weather that now reigned in these latitudes, and in preparation for the peculiarly active pursuits shortly to be anticipated, Perth, the begrimed, blistered old blacksmith, had not removed his portable forge to the hold again, after concluding his contributory work for Ahab's leg, but still retained it on deck, fast lashed to ringbolts by the foremast; being now almost incessantly invoked by the headsmen, and harpooneers, and bowsmen to do some little job for them; altering, or repairing, or new shaping their various weapons and boat furniture. Often he would be surrounded by an eager circle, all waiting to be served; holding boat-spades, pike-heads, harpoons, and lances, and jealously watching his every sooty movement, as he toiled. Nevertheless, this old man's was a patient hammer wielded by a patient arm. No murmur, no impatience, no petulance did come from him. Silent, slow, and solemn; bowing over still further his chronically broken back, he toiled away, as if toil were life itself, and the heavy beating of his hammer the heavy beating of his heart. And so it was.—Most miserable! A peculiar walk in this old man, a certain slight but painful appearing yawning in his gait, had at an early period of the voyage excited the curiosity of the mariners. And to the importunity of their persisted questionings he had finally given in; and so it came to pass that every one now knew the shameful story of his wretched fate. Belated, and not innocently, one bitter winter's midnight, on the road running between two country towns, the blacksmith half-stupidly felt the deadly numbness stealing over him, and sought refuge in a leaning, dilapidated barn. The issue was, the loss of the extremities of both feet. Out of this revelation, part by part, at last came out the four acts of the gladness, and the one long, and as yet uncatastrophied fifth act of the grief of his life's drama. He was an old man, who, at the age of nearly sixty, had postponedly encountered that thing in sorrow's technicals called ruin. He had been an artisan of famed excellence, and with plenty to do; owned a house and garden; embraced a youthful, daughter-like, loving wife, and three blithe, ruddy children; every Sunday went to a cheerful-looking church, planted in a grove. But one night, under cover of darkness, and further concealed in a most cunning disguise, a desperate burglar slid into his happy home, and robbed them all of everything. And darker yet to tell, the blacksmith himself did ignorantly conduct this burglar into his family's heart. It was the Bottle Conjuror! Upon the opening of that fatal cork, forth flew the fiend, and shrivelled up his home. Now, for prudent, most wise, and economic reasons, the blacksmith's shop was in the basement of his dwelling, but with a separate entrance to it; so that always had the young and loving healthy wife listened with no unhappy nervousness, but with vigorous pleasure, to the stout ringing of her young-armed old husband's hammer; whose reverberations, muffled by passing through the floors and walls, came up to her, not unsweetly, in her nursery; and so, to stout Labor's iron lullaby, the blacksmith's infants were rocked to slumber. Oh, woe on woe! Oh, Death, why canst thou not sometimes be timely? Hadst thou taken this old blacksmith to thyself ere his full ruin came upon him, then had the young widow had a delicious grief, and her orphans a truly venerable, legendary sire to dream of in their after years; and all of them a care-killing competency.

3.html是啥?

网页的骨架，用于负责哪里放文字，放图片，放按钮，放标题。

样子如图：

```
<!DOCTYPE html>
<html>
<body>
  <h1>我是标题</h1>
  <p>我是一段文字</p>
</body>
</html>
```

把这段代码复制到html文件：



三、API

1.什么是API? API有什么用?

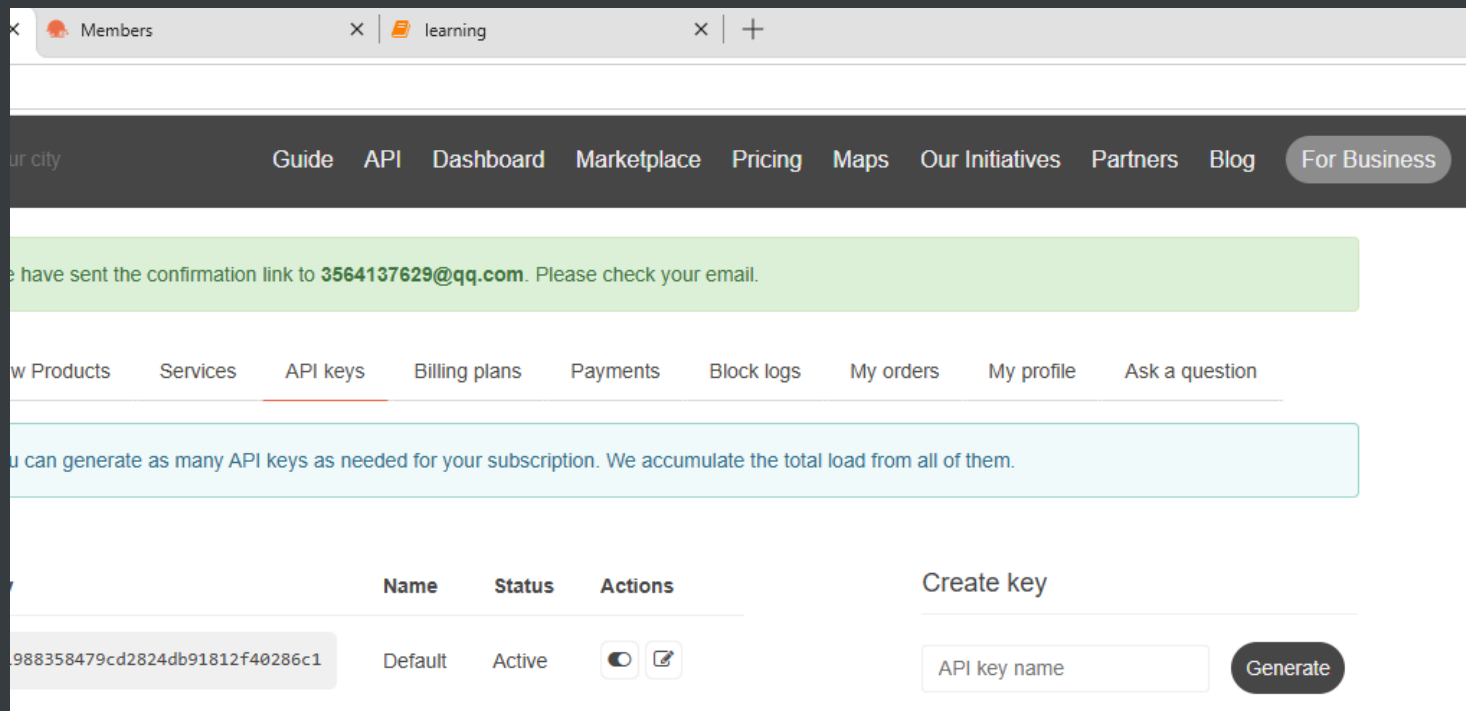
API是软件之间的接口，可以互相借助软件之间的数据、功能。

2.API key?

API key是一串是随机的字符，用于在调用API时来确认身份。

3.运用API在openweather网站获取天气信息

登录网站获取api key:



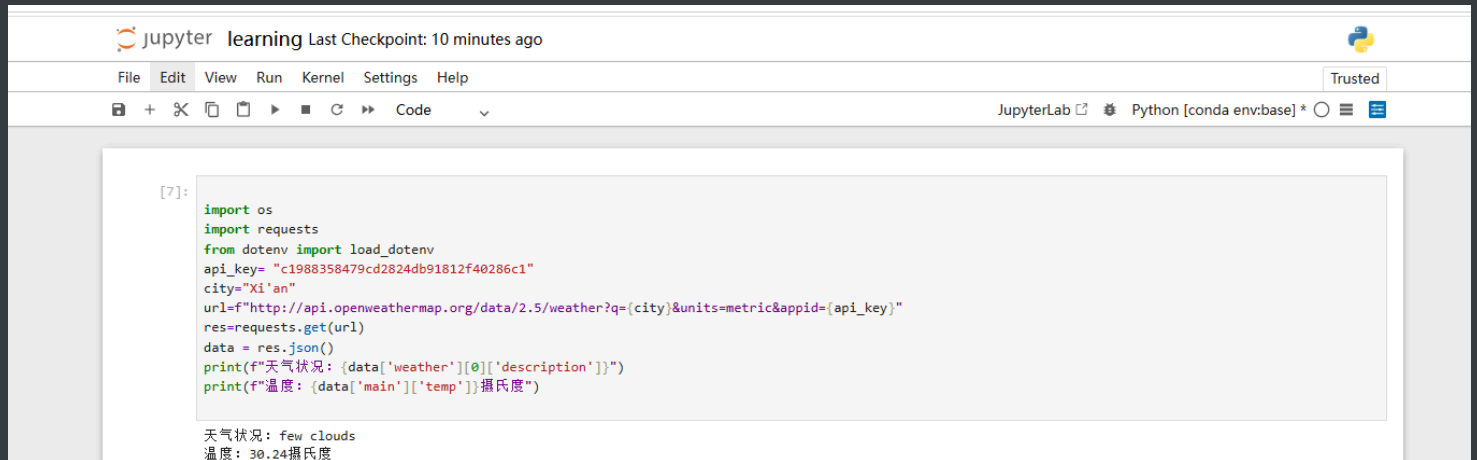
写获取信息的代码：

```
import requests #请求运用requests
api_key= "c1988358479cd2824db91812f40286c1"
city="Xi'an"
url=f"http://api.openweathermap.org/data/2.5/weather?q={city}&units=metric&appid={api_key}"

#api.openweathermap.org: 天气 API 服务器地址
#/data/2.5/weather: 获取 “当前天气” 的接口路径
#q={city}: 城市参数 (Xi'an)
#units=metric: 单位设置 → 摄氏度
#appid={api_key}: 把你的 API Key 填进去, 验证身份
res=requests.get(url)
data = res.json() #返回为json格式
"""
json 是这样, 就是字典的样子
{
  "weather": [{"description": "clear sky"}],
  "main": {"temp": 25}
}
"""
```

```
print(f"天气状况: {data['weather'][0]['description']}")  
print(f"温度: {data['main']['temp']}摄氏度")
```

运行后的样子:



The image shows a JupyterLab interface. The top bar includes the Jupyter logo, the name 'learning', and 'Last Checkpoint: 10 minutes ago'. Below this is a menu bar with 'File', 'Edit', 'View', 'Run', 'Kernel', 'Settings', and 'Help'. A 'Trusted' badge is visible on the right. The main area contains a code cell with the following Python code:

```
[7]:  
import os  
import requests  
from dotenv import load_dotenv  
api_key= "c1988358479cd2824db91812f40286c1"  
city="Xi'an"  
url=f"http://api.openweathermap.org/data/2.5/weather?q={city}&units=metric&appid={api_key}"  
res=requests.get(url)  
data = res.json()  
print(f"天气状况: {data['weather'][0]['description']}")  
print(f"温度: {data['main']['temp']}摄氏度")
```

Below the code cell, the output is displayed:

```
天气状况: few clouds  
温度: 30.24摄氏度
```